This is not a draft of my writeup, but an informal project update.
This project investigates the "Linear Representation Hypothesis" in the context of formal mathematics. Specifically, we ask if Large Language Models represent structural properties of Lean 4 tactic states as linear directions in their activation space. Cunningham et al. (2023) addresses the problem of polysemanticity by examining if features are entangled or reside in linear manifolds that we can interpret.

We use lean-dojo to extract tactic states from Mathlib and the Lean 4 standard library. Tactic states are labelled with a numerical value y representing the “Variable Count” in the local context. We extract internal activations, or the residual stream, using transformer_lens. A linear probe is then trained to map these vectors to our target labels. In future experiments we’ll look at a lot more than just Variable Count, but it’s a nice one for smoke tests because it’s easy to quickly and visually verify results.

We have successfully developed a pipeline that Extracts states and calculates variable counts using regex-based parsing, hooks into models like Pythia-160m to capture activations, and trains a basic linear MSE-loss probe. Having only tested on Pythia-160m, we don’t really see any remotely accurate guesses yet, but now that the infrastructure is in place training on models like deepseek-prover and goedel-prover to look at lean-4 trained models and llama as a text-trained model.

Bibliography:

- Cunningham, H., Ewart, A., Riggs, L., Huben, R., & Sharkey, L. (2023). *Sparse Autoencoders Find Highly Interpretable Features in Language Models.* arXiv preprint arXiv:2309.08600. (Foundational for addressing polysemanticity and identifying monosemantic directions in the residual stream).
- Park, J. S., et al. (2023). *Linear Representation Hypothesis.* (Theoretical basis for probing high-level concepts as specific directions in high-dimensional vector spaces).
- Yao, S., Yu, D., Zhao, J., Shafran, I., Griffiths, T. L., Cao, Y., & Narasimhan, K. (2023). *Tree of Thoughts: Deliberate Problem Solving with Large Language Models.* arXiv preprint arXiv:2305.10601. (Framework for strategic reasoning and lookahead in LLMs).
- Song, K., Tan, Y., Zhang, L., Jiang, Z. J., Henein, S., & Ganeshan, A. (2024). *Lean Copilot: Large Language Models as Copilots for Theorem Proving in Lean.* arXiv preprint arXiv:2404.12534. (Provides the interface for running LLM inference natively within Lean 4).
- Yang, K., Swope, A. M., Gu, A., Chalamala, H., Song, P., Yu, S., ... & Klein, D. (2023). *LeanDojo: Theorem Proving with Retrieval-Augmented Language Models.* arXiv preprint arXiv:2306.15626. (The data extraction framework used to interact with Mathlib and Std4 repositories).
- Fang, Y., Huang, S., Yu, X., et al. (2025). *NL2Lean: Translating Natural Language into Lean 4 through Multi-Aspect Reinforcement Learning.* EMNLP 2025. (Discussion of structural and semantic alignment in formalization tasks).